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Time-Varying Complementary Filtering for Attitude Estimation

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Abstract— **Complementary filtering (CF) is a well known method that can effectively fuse a gyroscope and accelerometer measurement in order to robustly estimate the attitude of a rigid body in a planar single degree of freedom (DOF) setting. The attitude can be estimated individually by either integrating the gyroscope measurement or by calculating the inverse tangent of the components of a 2-axis accelerometer. The gyroscope can adequately estimate the angle in the higher frequency region, but suffers from drift issues at low frequency, whereas the accelerometer can accurately measure the acceleration and thus direction of gravity, but loses this accuracy when faced with motion accelerations. CF traditionally uses linear time invariant filters, however, this paper presents an extension to the CF method by proposing time-varying parameters. A fuzzy logic method is developed to adjust the parameters. Stability analysis as well as experimental results are presented to verify the proposed method.**

I. INTRODUCTION

As robotic systems are becoming more mobile, the requirement for improved quality of attitude sensing is becoming more necessary, where attitude is defined as three-dimensional orientation. The main challenge of attitude estimation objective stems from the limitations of sensors. If a mobile robot is physically connected to the ground by linkages, e.g., humanoid robot legs during stance phase, then estimation of attitude for a mobile robot is possible with the use of encoders. However, if the robot is traversing complex terrain and is continuously being perturbed from the ground, then attitude estimation becomes less trivial. Many sensors have been studied for this purpose; for example, inertial measurement units (IMUs), indoor/outdoor GPS, magnetometers, vision, ultrasonic senors or range finders [1], [2], [3], [4]. However, in the specific case where only the single degree of freedom (DOF) pitch or roll attitude with respect to the horizon are to be estimated, accelerometers and rate gyroscopes are well suited for this purpose. Moreover, gyroscopes and accelerometers are appropriate for mobile applications because they are light weight, provide fast update rates, require low computation, and are low in cost. Since accelerometers can detect gravity, they can be utilized to estimate the direction of gravity when stationary. On the other hand, gyroscopes measure body angular rates and make it possible to estimate the attitude by integrating the signal. From these two sensing methods the same physical quantity, pitch or roll, can be estimated. Since each sensor provides different frequency domain characteristics, an improved result is expected if the two methods are fused intelligently.

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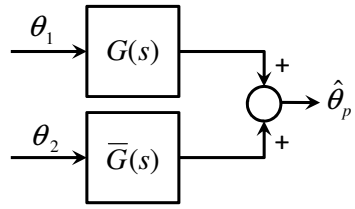

Fig. 1: Block diagram of general two input complementary filtering.

In order to obtain a better estimate of the physical quantity from fusing multiple sensing methods, various signal processing algorithms have been studied. A kinematic Kalman filter (KKF) can fuse sensors which are kinematically related. For example, the acceleration measured by an accelerometer can be related to the position measured by an encoder [5], [6], so that a mathematical model between the two signals (i.e., a double integrator) can be established. In particular, the KKF has shown good performance when estimating the position and velocity of a robotic arm's end-effector [7]. The design of the KKF requires knowledge of the type of noise characteristics, and it is optimal only when the noise is zero mean, white, and Gaussian distributed. However, the noise in practice often exhibits more complicated characteristics and in some situations be best be described by qualitative means.

Complementary filtering (CF), as shown in Fig. 1, is an alternative signal processing method that can be used for the same purpose [8], [9], [10]. The CF method has been widely applied to flying systems [11] and terrestrial robotics [12] due to its uncomplicated structure and effectiveness [10]. Also, when compared to the steady state Kalman filter, CF has shown to be equivalent while requiring less computation [13]. In particular, the CF method is effective when the same physical quantity can be estimated from multiple sensing methods with different frequency characteristics.

The conventional CF method utilizes linear-time-invariant (LTI) filters with different frequency characteristics, such that only the reliable components are selectively extracted in the frequency domain. Since integration is required to estimate the pitch or roll attitude from a rate gyroscope, where the integration amplifies uncertainty at low frequency, the reliability of the estimate by gyroscope is poor at low frequency (e.g., drift). The limitation in estimating attitude from accelerometers is evident due to the difficulty in decoupling the gravity vector from the higher frequency

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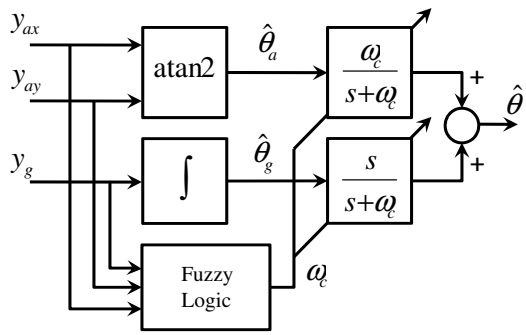

Fig. 2: Block diagram of proposed TVCF method with gyroscope and accelerometer signals.

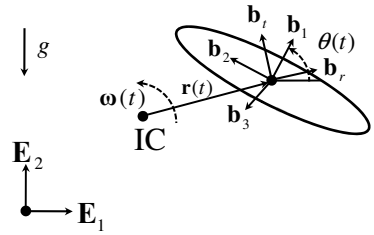

Fig. 3: General diagram of a planar rigid body.

motion accelerations, and thus its reliability is poor at high frequency. Therefore, the CF method can be applied to this problem, where a lowpass filter and a highpass filter are used as the LTI filters, $G(s)$ and $\bar{G}(s)$, in Fig. 1. However, the challenge of the CF method is to design the LTI filters according to the frequency dependent reliabilities. When the frequency characteristics are time-invariant and clearly distinguished, the design of the filters require minor tuning. However, in many cases, the frequency characteristics are not fixed and the simple LTI filters may not produce the best result.

In this paper, a time-varying complementary filtering (TVCF) method is proposed as shown in Fig. 2. The proposed method utilizes linear time-varying (LTV) filters for estimating a physical quantity from multiple sensors whose reliability in the frequency domain is time-varying. The advantage of this method is that it improves the performance of the CF method with more degrees of freedom in the design of the filters. The main issue is determining the best conditions with which to adapt the LTV filters. A fuzzy logic scheme is proposed to formalize the qualitative conditions for adaptation. The performance is verified against encoder measurements from a 3DOF apparatus that allows for instantaneous rotations in the $\mathbf{E}_1 - \mathbf{E}_2$ plane as opposed to a fixed pivot.

This paper is organized as follows. In section II, the standard CF method will be reviewed. In section III the details of the TVCF will be explained along with a proof of stability. Then, in section IV, the experimental results will be discussed displaying the benefit of the method and finally in section V, the conclusions will be drawn.

II. Previous Works: CF Method

The main idea of the CF method, represented in Fig. 1, is to fuse two measurements, $\theta_1(t)$ and $\theta_2(t)$, of the same physical quantity, $\theta_p(t)$. $\bar{G}(s)$ is often designed as $1-G(s)$, such that $G(s)+\bar{G}(s)=1$. In this case, the estimate $\hat{\theta}_p(t)$ is exactly the physical quantity if the measurements are without noise and error.

Suppose that $\theta_1(t)$ and $\theta_2(t)$ are reliable at low and high frequencies respectively. In a first order implementation, $G(s)$ can be set as a low pass filter with a fixed cut off frequency, ω_c, i.e.

$$G(s) = \frac{\omega_c}{s+\omega_c} \tag{1}$$

The complement of $G(s)$, i.e., $\bar{G}(s) = 1 - G(s)$, is then

$$\bar{G}(s) = \frac{s}{s+\omega_c} \tag{2}$$

which becomes a high pass filter. Note that $G(s)+\bar{G}(s)=1$ for all s and consequently there is no phase delay in the estimate, $\hat{\theta}_p(t)$.

If $\theta_1(t) = \theta_p(t) + \eta_1(t)$ and $\theta_2(t) = \theta_p(t) + \eta_2(t)$, where η_1 is high frequency noise and η_2 is low frequency noise both relative to ω_c, then by (1, 2) the estimate by CF is

$$\hat{\theta}_p = G(s)[\theta_p + \eta_1] + \bar{G}(s)[\theta_p + \eta_2] \tag{3}$$

$$= \theta_p + G(s)[\eta_1] + \bar{G}(s)[\eta_2]. \tag{4}$$

where the signals are in the Laplace domain. If the cut-off frequency, ω_c, is properly selected, the magnitude of $G(s)[\eta_1] + \bar{G}(s)[\eta_2]$ in the time domain is small compared to θ_p due to the characteristics of the noise and LTI filters.

With the goal of estimating the pitch or roll attitude, $\theta(t)$, which will now be referred to as just attitude, of a rigid body as shown in Fig. 3, three measurements from a single axis gyroscope and a dual axis accelerometer are used and represented as

$$y_g(t) = \boldsymbol{\omega}(t) \cdot \mathbf{b}_3 + \beta_g(t) + \eta_g(t) \tag{5}$$

$$y_{ax}(t) = \mathbf{a}_m(t) \cdot \mathbf{b}_1 + \mathbf{g} \cdot \mathbf{b}_1 + \beta_{ax}(t) + \eta_{ax}(t) \tag{6}$$

$$y_{ay}(t) = \mathbf{a}_m(t) \cdot \mathbf{b}_2 + \mathbf{g} \cdot \mathbf{b}_2 + \beta_{ay}(t) + \eta_{ay}(t) \tag{7}$$

where $\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{b}_3$, are body frame coordinates perpendicular to each other. The gyroscope measurement, $y_g(t)$, aligned along $\mathbf{b}_3$, measures the angular rate, $\boldsymbol{\omega}(t) = \dot{\theta}(t)\mathbf{b}_3$, but also contains bias, $\beta_g(t)$, and noise, $\eta_g(t)$. The two orthogonal accelerometer measurements, $y_{ax}(t)$ and $y_{ay}(t)$, are aligned along the axes, $\mathbf{b}_1$ and $\mathbf{b}_2$, respectively. They measure the body fixed basis components of the gravity vector, $\mathbf{g} \cdot \mathbf{b}_1$ and $\mathbf{g} \cdot \mathbf{b}_2$, where $\mathbf{g} = -g\mathbf{E}_2$, and the motion accelerations, $\mathbf{a}_m(t) \cdot \mathbf{b}_1$ and $\mathbf{a}_m(t) \cdot \mathbf{b}_2$. The motion accelerations are assumed to be much more significant than the bias or noise in the accelerometer sensors.

With a known initial condition, θ_o, the single axis gyroscope can be integrated to achieve an attitude estimate, $\hat{\theta}_g(t) = \int_0^t y_g(\tau)d\tau + \theta_o$. On the other hand, in the near static or constant velocity condition, the motion accelerations, $\mathbf{a}_m(t)$ in (6, 7), are near zero and the orthogonal components

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TABLE I: Rule Basis of Fuzzy Logic.

Condition	$\lvert\lVert \mathbf{y}_a(t)\rVert_2 - g\rvert$	$\lVert \dot{\mathbf{y}}_a(t)\rVert_2$	$\lvert y_g(t)\rvert$	$\lvert \dot{y}_g(t)\rvert$	$\mu(t)$
stationary or no rotation & constant linear velocity	*low*	*low*	*low*	*low*	1
nonstationary, dashed circle ($\lVert y_a\rVert_2 = g$)	N/A	*high*	N/A	N/A	0
linear acceleration	*high*	N/A	N/A	N/A	0
centripetal acceleration	N/A	N/A	*high*	N/A	0
centrifugal acceleration	N/A	N/A	N/A	*high*	0

of the dual axis accelerometer can estimate the attitude from the gravity vector by $\hat{\theta}_a(t) = \arctan 2(y_{ax}(t), y_{ay}(t))$. The $\arctan 2()$ function is chosen as opposed to the $\arctan()$ function to estimate $\theta(t)$, since the range is $[-\pi, \pi)$ (rad) and not just $(-\frac{\pi}{2}, \frac{\pi}{2})$ (rad).

Individually, the gyroscope gives a decent estimate of angle at higher frequencies, but this estimate tends to drift due to the bias and noise. The accelerometer gives an accurate static estimate, but this estimate is predominately corrupted by motion accelerations. Therefore, the attitude estimates from the gyroscope and accelerometer can be modeled as

$$\hat{\theta}_g(t) = \theta(t) + \eta_d(t) \tag{8}$$

$$\hat{\theta}_a(t) = \theta(t) + \eta_m(t) \tag{9}$$

where $\theta(t)$ is the true attitude angle to be estimated by CF, and $\eta_d(t)$ and $\eta_m(t)$ are the measurement errors corresponding to the drift and motion acceleration terms respectively. In order to apply the CF method, $\eta_d(t)$ and $\eta_m(t)$ are treated as noise; $\eta_d(t)$ is low frequency and $\eta_m(t)$ is high frequency. Using this method a more accurate estimate of $\theta(t)$ compared to the individual sensors can be produced. Choosing a larger ω_c places more trust on the accelerometer estimate, $\hat{\theta}_a(t)$, while choosing a smaller ω_c places more trust on the gyroscope estimate, $\hat{\theta}_g(t)$.

III. TIME VARYING COMPLEMENTARY FILTERING

During general planar motion, the rigid body is undergoing translation and rotation about an instantaneous center, which is designated by (IC) as shown in Fig. 3. The motion acceleration in (6, 7) constrained to the $\mathbf{E}_1 - \mathbf{E}_2$ plane can be written as

$$\mathbf{a}_m = \mathbf{a}_t + \boldsymbol{\omega} \times [\boldsymbol{\omega} \times \mathbf{r}] + \boldsymbol{\alpha} \times \mathbf{r} \tag{10}$$

where $\mathbf{a}_t$ is the translational acceleration and $\mathbf{r} = r\mathbf{b}_r$ is the radial vector. Although, $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ in (10) are arbitrary vectors in three dimensional space, it is assumed that $\boldsymbol{\omega} = \dot{\theta}\mathbf{b}_3$ and $\boldsymbol{\alpha} = \ddot{\theta}\mathbf{b}_3$, for the sake of simplicity. The last two terms constitute the centripetal and tangential acceleration. Ideally, if the motion accelerations, $\mathbf{a}_m$, could be measured or estimated, then they could be decoupled from the gravity vector, $\mathbf{g}$, in (6) and (7). This would allow for the accelerometer estimate to be trusted for a higher frequency range. However, the translational acceleration, $\mathbf{a}_t$, and the instantaneous center, which determines the the magnitude and direction of the radial vector, $\mathbf{r}$, cannot be directly

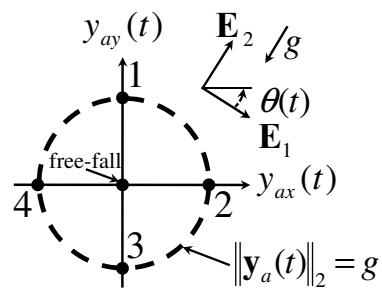

Fig. 4: Circle of ambiguity from the accelerometer sensors due to coupling of motion accelerations and gravitational field.

measured. Therefore, if *when* the motion accelerations occur can be detected, then the cut off frequency, ω_c in (1) and (2), can be adjusted to a low value, ω_{low}, and effectively filter out the high frequency noise, $\eta_m(t)$, in (9). Conversely, when the rigid body is not accelerating, the cutoff frequency can be increased to a high value, ω_{high}, to update the estimate from the accelerometer, thus proposing the Time Varying Complementary Filter (TVCF).

A. Fuzzy Logic

In order to formalize our engineering intuition we chose a fuzzy logic method to discern the non-accelerating state. A simple rule-base to determine the cut-off frequency is described in Table I. The last column contains the output of the fuzzy logic, μ, which is valued in the range of [0,1] and signifies how to transition between ω_{high} or ω_{low} as

$$\omega_c(t) = \mu(t)\omega_{high} + (1 - \mu(t))\omega_{low} \tag{11}$$

Notice that if $\mu = 1$ then $\omega_c = \omega_{high}$ thus signifying when the estimate from the accelerometer is more trustworthy. Inputs to the fuzzy logic, which determine the value of μ, can be classified qualitatively as either *low* or *high*. A logistic function will later be explained to convert this qualitative descriptions to quantitative logic values.

From (10) the centripetal and tangential accelerations occur when the angular rate, $\dot{\theta}(t)$, and the angular acceleration, $\ddot{\theta}(t)$, are *high*. This information can be determined from the magnitude of the gyroscope signal, $|y_g(t)|$, and the signals time derivative, $|\dot{y}_g(t)|$, which is represented in the last two rows of Table I.

The rigid body is assumed to exist in a gravitational field of $-1g$ in the $\mathbf{E}_2$ world frame direction. The accelerometer

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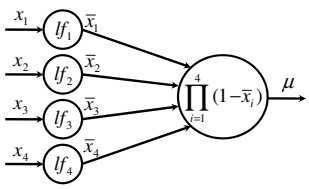

Fig. 5: Network diagram of fuzzy logic where lf_i represents the logistic function in Fig. 6

.

measurement vector is first defined as $\mathbf{y}_a(t) = [y_{ax}(t)\ y_{ay}(t)]^T$. Any translational acceleration will occur when the magnitude of the accelerometer measurement vector, $\|\mathbf{y}_a(t)\|_2 = \sqrt{y_{ax}(t)^2 + y_{ay}(t)^2}$, is not close to the gravitational acceleration, g. One example is free-fall; $\|\mathbf{y}_a(t)\|_2 = 0$. Therefore, the estimate by accelerometer is not trustable if $|\|\mathbf{y}_a(t)\|_2 - g|$ is *high*. This logic is represented by row three in Table I.

The equation, $\|\mathbf{y}_a(t)\|_2 = g$, represented by the dashed circle of ambiguity in Fig. 4, holds the exact point of non-acceleration and also contains points of acceleration and shows the limitation of only checking $\|\mathbf{y}_a(t)\|_2$. This fact is apparent if one briefly assumes that the sensor signals, $y_{ax}(t)$ and $y_{ay}(t)$, are aligned with the $\mathbf{E}_1$ and $\mathbf{E}_2$ axes respectively, then the non-acceleration point of interest is point 1, which cannot be distinguished from, falling down at $2g$ (point 3), or accelerating perfectly horizontal and down at $1g$ (point 2 and 4). Since the dashed circle is not stable in any sense, other than possibly the non-acceleration state, it would be improbable to stay along the circle for a prolonged period of time. Therefore, with this assumption, acceleration can be discerned from non-acceleration even when $|\|\mathbf{y}_a(t)\|_2 - g|$, $|y_g(t)|$, and $|\dot{y}_g(t)|$ are *low* by checking if the magnitude of the *jerk*, $\|\dot{\mathbf{y}}_a(t)\|_2 = \sqrt{\dot{y}_{ax}(t)^2 + \dot{y}_{ay}(t)^2}$, is *high*. This is represented in the first three rows of Table I.

With the logic built, the inputs to the fuzzy logic, x_i, diagrammed in Fig. 5 can be selected from the column headers in Table I as

$$x_1(t) = |\|\mathbf{y}_a(t)\|_2 - g| \tag{12}$$
$$x_2(t) = \|\dot{\mathbf{y}}_a(t)\|_2 \tag{13}$$
$$x_3(t) = |y_g(t)| \tag{14}$$
$$x_4(t) = |\dot{y}_g(t)| \tag{15}$$

These inputs are then scaled by a logistic function, lf_i, to a quantitative unitless output, $\bar{x}_i$, with range of (0,1) representing the *low* and *high* descriptions in the logic table. Although any logic function can be used, a hyperbolic tangent function

$$\bar{x}_i(t) = 0.5(1 + \tanh(s_i(x_i(t) - x_{oi}))) \tag{16}$$

was chosen for its smooth transition properties shown by Fig. 6. Here, x_{oi} represents the 50 % transition point or when the value of the input, x_i, corresponds to an output, $\bar{x}_i = 0.50$. The value s_i represents the slope of transition, for example larger values of s_i corresponds to a sharper transition.

From inspection of the logic table it is clear that only when *all* the inputs are low it is valid to assume a non-accelerating

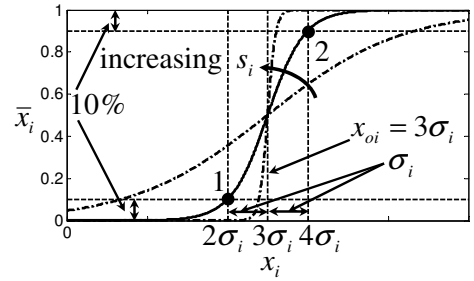

Fig. 6: Logistic function, lf_i, of (16) displaying the result of increasing the slope s_i. The intersections of the 10 % lines and σ_i lines at points 1 and 2 designate the design points for calculation of s_i.

state and to choose $\mu = 1$. Therefore, the rule basis of the fuzzy logic in Table I can be formulated by a multiplication function, i.e.,

$$\mu(t) = \prod_{i=1}^{4}(1 - \bar{x}_i(t)) \tag{17}$$

Notice that (17) returns one only when all $\bar{x}_i = 0$, which follows the rules in Table I.

B. Stability Analysis

In the previous section, a continuous time representation of the TVCF was presented. Since the TVCF must be implemented in the discrete time domain, the stability of TVCF is analyzed in the discrete time domain. Although many continuous to discrete time transformations are available [14], *Tustin's* bilinear transformation, was chosen so that the LTV filter relationship, $G(z) + \bar{G}(z) = 1$, could be maintained where z^{-1} is a step delay operator. If the same input signal, u, is assumed to enter both LTV filters and y_{lp} and y_{hp}, are the outputs of $G(z)[u]$ and $\bar{G}(z)[u]$ respectively then the resulting difference equations are

$$\begin{aligned} y_{lp}(k) = &-\frac{-2+\omega_c(k)dt}{2+\omega_c(k)dt} y_{lp}(k-1) \\ &+ \frac{\omega_c(k)dt}{2+\omega_c(k)dt} u(k) + \frac{\omega_c(k)dt}{2+\omega_c(k)dt} u(k-1) \end{aligned} \tag{18}$$

$$\begin{aligned} y_{hp}(k) = &-\frac{-2+\omega_c(k)dt}{2+\omega_c(k)dt} y_{hp}(k-1) \\ &+ \frac{2}{2+\omega_c(k)dt} u(k) - \frac{2}{2+\omega_c(k)dt} u(k-1) \end{aligned} \tag{19}$$

where dt is the sampling time. The output, $y(k)$, of the TVCF is calculated by adding (18) and (19), which results in

$$\begin{aligned} y(k) &= y_{lp}(k) + y_{hp}(k) \\ &= -\rho(k)y(k-1) + \rho(k)u(k-1) + u(k) \end{aligned} \tag{20}$$

where

$$\rho(k) = \frac{-2+\omega_c(k)dt}{2+\omega_c(k)dt} \tag{21}$$

In the case where $\rho(k)$ is constant, it can be easily proven that the discrete time transfer function relating $u(k)$ to $y(k)$ obtained from (20) becomes one and that $y(k) = u(k)$ for $k \neq 0$. To prove the error convergence in the time-varying case

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the error is first defined as $e(k) = y(k) - u(k)$. By rearranging (20), the error dynamics is obtained as

$$e(k) = -\rho(k)e(k-1) \tag{22}$$

For a given initial condition, $e(0)$, the solution of (22) is

$$e(k) = [\prod_{i=1}^{k} -\rho(i)]e(0) \tag{23}$$

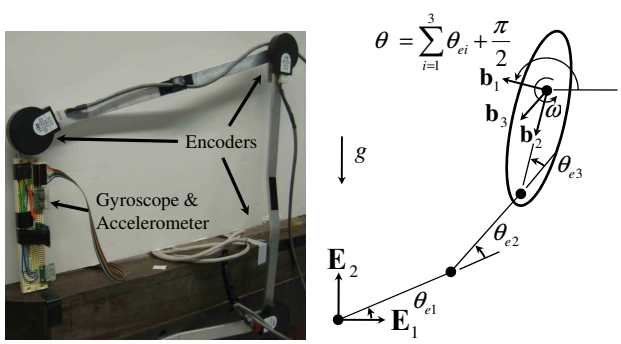
If $|\rho(k)| < 1$ for all k then the output of the TVCF, $y(k)$, converges to $u(k)$ monotonically even when $e(0) \neq 0$. Note in (21) that $|\rho(k)| < 1$ as long as $\omega_c(k)dt > 0$. Therefore, the stability of the TVCF is guaranteed if $\omega_c(k) > 0$ for all k since the sampling time is always $dt > 0$.

IV. EXPERIMENTAL RESULTS

A. Experimental Apparatus

For verification of the proposed TVCF method, the true attitude angle should be measured by an accurate measurement system. In this paper, a linkage system equipped with high resolution encoders was set up for this purpose as shown in Fig. 7a. The linkage system was designed with 3DOF to allow for rotation and translation in the $\mathbf{E}_1 - \mathbf{E}_2$ plane as shown in Fig. 7b. The true attitude, $\theta(t)$, was calculated from the sum of three high resolution joint encoder measurements, $\theta_{ei}(t)$, as, $\theta(t) = \sum_{i=1}^{3} \theta_{ei}(t) + \frac{\pi}{2}$.

The algorithms and data collection were implemented on National Instrument's 7831R FPGA and LabVIEW with a fixed sampling time of 0.005 (s). An Invensense, IDG-650, ± 2000 $(\frac{deg}{s})$, gyroscope and an Analog Devices, ADXL203, ± 1.7 g, dual axis accelerometer were mounted on the third link at the origin of the body fixed basis in Fig. 7b. For verification a US Digital, E6, 2048 count per revolution, optical encoder was mounted at each joint. In each experimental trial, the link was randomly hand shaken and rotated while also providing short periods of rest.



(a) Photograph of experimental apparatus.

(b) Schematic of experimental apparatus.

Fig. 7: Experimental apparatus

B. Selection of Parameters

The TVCF method requires the selection of ten parameters; ω_{low}, ω_{high}, and all of the x_{oi} and s_i values. The following can serve as a guideline for choosing them. The parameters x_{oi} and s_i determine the non-accelerating conditions. If the input signals, $\bar{x}_i$, are assumed to be zero mean and normally distributed when the system is static, the parameters can be conveniently calculated from one specification; the standard deviation, σ_i, of the noise in each input signal, $\bar{x}_i$. The 50 % transition point, x_{oi}, can be set equal to $3\sigma_i$ to encapsulate 99.7 % of the input signals in a statistical sense when they are static. The slope, s_i, can be calculated from the intersection of when x_i is one σ_i away from x_{oi} and when $\bar{x}_i(t)$ is equal to 10 % of the the total distance from either bound of $\bar{x}_i(t)$. This is evident from point 1 and 2 in Fig. 6. With these design points, the slope can be calculated from (16) as

$$s_i = \frac{\tanh^{-1}(-0.8)}{-\sigma_i} \tag{24}$$

Sensor bias significantly affects the performance of detecting non-acceleration in the signals, $\bar{x}_1$ and $\bar{x}_3$, but does not affect $\bar{x}_2$ and $\bar{x}_4$ since they are differentiated signals. Therefore, multiplying σ_1 and σ_3 by a factor, $\gamma_i > 1$, $i = 1, 3$, produces larger fictitious standard deviations that can account for the upper limit of the sensor bias. By inspection of the signals during static conditions, a value of $\gamma_i = 5$ allowed capture of the upper limit of the sensor bias. Both the sensor bias and standard deviation of the signal noise can be determined experimentally or calculated from the sensor specifications.

The parameter, ω_{low}, should be chosen low enough to ignore the accelerometer during motion. A preselected value of $\omega_{low} = 0.01$ $(\frac{rad}{s})$ is appropriate. When the algorithm decides to trust the accelerometer estimate, the cutoff frequency adjusts towards ω_{high}. Also at this time, the attitude estimate may be affected by drift, but converges faster to the accelerometer estimate with a larger value of ω_{high}. The upper limit of ω_{high} should conservatively be no more than $1/25$ of the sampling rate; $\omega_{high} < 50$ $(\frac{rad}{s})$ for a sampling rate of 0.005 (s) [15]. A preselected value of $\omega_{high} = 10$ $(\frac{rad}{s})$ was found appropriate. In the Section IV-D, the adequacy of these two parameters will be justified. The preselected set of parameters are shown in the second column of Table II.

C. Experimental Results

Experiments measuring the attitude estimates from the accelerometer, gyroscope, CF method and TVCF method were performed. For each estimation method, ten, one minute long, trials were taken. The experimental error was also recorded and defined by the difference between the true angle and the estimate of the various methods as $e_{exp}(k) = \theta(k) - \hat{\theta}(k)$. To analyze the performance of each trial the root-mean-square (rms) of the experimental error, ξ, defined by

$$\xi = \sqrt{\frac{\sum_{k=1}^{n} e_{exp}^2(k)}{n}} \tag{25}$$

where n is the total number of samples, was calculated. The ten ξ values for each method are plotted in an errorbar plot as shown in Fig. 8. The attitude estimate from the

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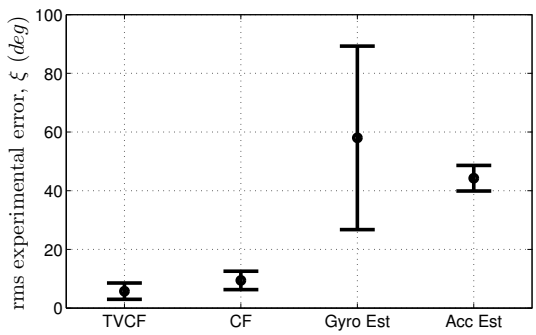

Fig. 8: Errorbar plot comparing the rms experimental error, ξ, for each of the different methods. The center circle in each category represents the mean value while the error bars denote one standard deviation.

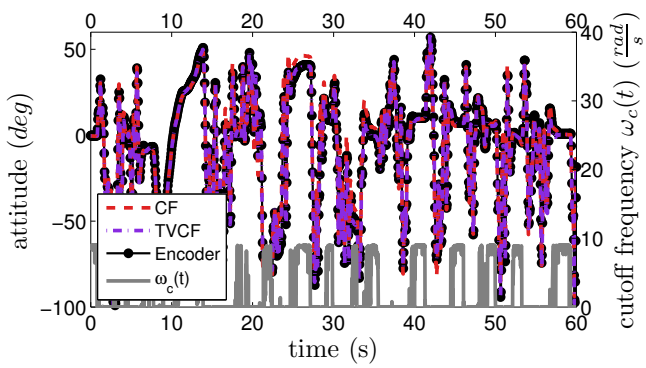

Fig. 9: Representative plot of the estimated attitude for each of the different methods as well as the corresponding $\omega_c(t)$ for the TVCF.

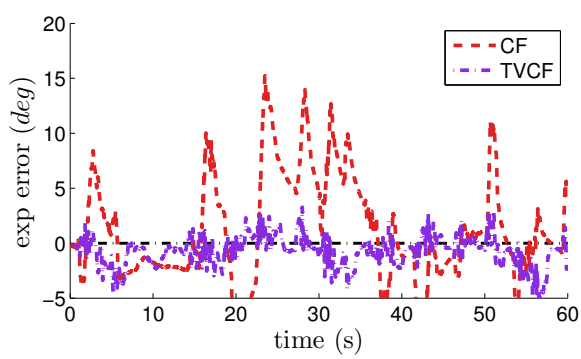

Fig. 10: Representative plot of the experimental error, e_{exp}, for each of the different methods.

accelerometers has a mean ξ of approximately 45° with low variance, while the gyroscope estimate has high variance and a mean ξ of approximately 60°. The attitude estimate from the CF method has a mean ξ of approximately 10° while the mean ξ for the proposed TVCF method is approximately 5°, which is a significant improvement (*t-test*, $p = 0.0124 < 0.05$) [16]. A probability value, $p < 0.05$, indicates that there is a significant difference between the mean of the two sets of ξ.

Fig. 9 and Fig. 10 are representative experimental attitude

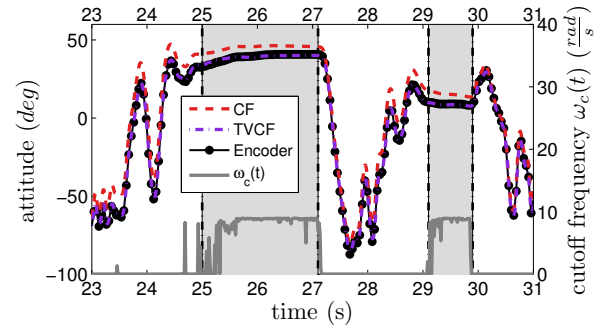

Fig. 11: Zoomed in plot of Fig. 9. Grey areas represent periods of rest.

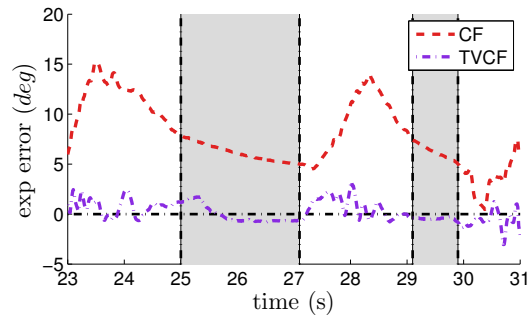

Fig. 12: Zoomed in plot of Fig. 10. Grey areas represent periods of rest.

TABLE II: Parameters.

Parameter	Initial Value	Optimized Value
ω_{high}	10 $(\frac{rad}{s})$	9.1 $(\frac{rad}{s})$
ω_{low}	0.01 $(\frac{rad}{s})$	0.024 $(\frac{rad}{s})$
σ_1	0.0017 (g)	2.56 (g)
σ_2	0.004 $(\frac{dg}{dt})$	1.4 $(\frac{dg}{dt})$
σ_3	1.2 $(\frac{deg}{s})$	5.8 $(\frac{deg}{s})$
σ_4	320 $(\frac{deg}{s^2})$	320 $(\frac{deg}{s^2})$

and error plots of one of the ten experiments for the CF and TVCF methods. They are taken over the same trials and time periods respectively. Both, the conventional CF method and TVCF match the true angle measured by the encoder fairly well. Since the gyroscope estimate drifts, both estimates will suffer from drift during periods of movement. However, during periods of rest denoted by the grey areas in Fig. 11 and Fig. 12, the TVCF is able to adapt the complementary filters cutoff frequency to high and the estimate can quickly converge to the true attitude faster than the CF method.

D. Adequacy of the Selected Parameters

The adequacy of selecting the values for just ω_{high} and ω_{low} was first analyzed. Ten, one minute trials recording the data from the encoder, gyroscope, and accelerometer were collected and the TVCF algorithm was implemented off-line. The values of ω_{high} and ω_{low} were varied while the precalculated logistic parameters, x_{oi} and s_i, were fixed. The median rms errors between the encoder data and the off-line

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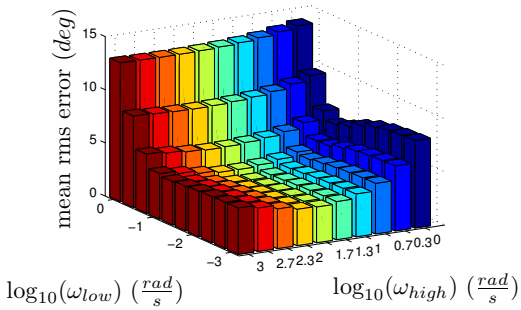

Fig. 13: Three dimensional plot of the TVCF method performance for varying values of ω_{high} and ω_{low}. Each column represents the mean rms error over ten, one minute trials. For the preselected parameters in the previous section, the performance sensitivity is quite low due to the flat geometry at the bottom of the basin.

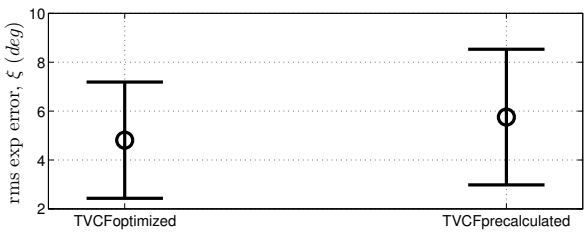

Fig. 14: Errorbar plot comparing the rms experimental error, ξ, of the TVCF with optimized and precalculated parameters. The center circle in each category represents the mean value while the error bars denote one standard deviation.

TVCF estimate similarly defined in (25) were plotted in Fig. 13. Notice the median rms error for the preselected values, $\omega_{high} = 10$ $(\frac{rad}{s})$ and $\omega_{low} = 0.1$ $(\frac{rad}{s})$ mentioned in section IV-B, fall near the minimum of the plot. Also, the median rms errors surrounding the preselected values deviate little and further give confidence in their selection if the TVCF method were to be implemented on other systems.

Additionally, to test the adequacy of all the parameters, ω_{high}, ω_{low}, x_{oi}, and s_i, an off-line optimization procedure obtained optimized values using the preselected values as the initial parameters. Again, the TVCF algorithm was implemented off-line with the same data. The *Matlab* function *fmincon* was employed for the off-line optimization procedure and minimized the same rms of the error. As expected, the optimization produced ten varying sets of values, which were then averaged into one final set of optimized parameters as shown in the right column of Table II. The real time performance using the precalculated parameters were then tested against the performance using the optimized values. As expected, the TVCF with optimized parameters showed a slight decrease in the median experimental error as shown in Fig. 14, but the improvement was negligible (*t-test*, $p = 0.4252 > 0.05$) also indicating a low sensitivity. This result supports the adequacy of the aforementioned guidelines since any verification by encoder data would most likely not be available in a practical setting.

V. CONCLUSION

In this paper, a new approach for a single DOF attitude estimation is proposed using a TVCF method. Significant improvements were achieved over the conventional CF method while still maintaining ease of implementation. The effectiveness has been verified with experimental results. This work reaches a general set of applications that require effective yet tractable attitude estimates in pitch and roll.

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